



A non-commutative algorithm for multiplying a 3x4 matrix by a 4x7 matrix using 63 non-complex multiplications

Jean-Guillaume Dumas, Clément Pernet, Alexandre Sedoglavic

► To cite this version:

Jean-Guillaume Dumas, Clément Pernet, Alexandre Sedoglavic. A non-commutative algorithm for multiplying a 3x4 matrix by a 4x7 matrix using 63 non-complex multiplications. 2025. <hal-05121550v2>

HAL Id: hal-05121550

<https://hal.science/hal-05121550v2>

Preprint submitted on 15 Jul 2025

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

A non-commutative algorithm for multiplying a 3×4 matrix by a 4×7 matrix using 63 non-complex multiplications

Jean-Guillaume Dumas,

Jean-Guillaume.Dumas@univ-grenoble-alpes.fr

Univ. Grenoble Alpes, CNRS, Grenoble INP – UGA, LJK,
38000 Grenoble, France

Clément Pernet,

Clement.Pernet@univ-grenoble-alpes.fr

Univ. Grenoble Alpes, CNRS, Grenoble INP – UGA, LJK,
38000 Grenoble, France

Alexandre Sedoglavic,

Alexandre.Sedoglavic@univ-lille.fr

Univ. Lille, CNRS, UMR 9189 CRISTAL,
F-59000 Lille, France

July 15, 2025

Abstract

The number of scalar multiplications required to multiply a 3×4 matrix by a 4×7 matrix was reduced from 66 in [12] to 63 in [10] by an algorithm over complex numbers.

We propose an algorithm in 63 multiplications with only rational coefficients, hence removing the requirement that the base field for the arithmetic operations contains a square root of -1 , while it still needs to contain an inverse of 2. It was derived from the latter one, under the action of an isotropy which happen to project the algorithm on the field of rational numbers. We also produce a straight line program of this algorithm.

1 Introduction

V. Strassen introduced in [13] the first sub-cubic time algorithm for matrix product, with 7 recursive multiplications and, thus, a complexity of $O(n^{2.81})$.

Recently, an algorithm to multiply a 3×4 matrix by a 4×7 requiring only 63 scalar complex-valued multiplications was found using a pipeline of large language models orchestrated by an evolutionary coding agent [10].

We present in this note a *equivalent variant* of this algorithm that only use rational coefficients and no longer rely on complex values: in fact, we consider the *tensor decomposition representation* encoding the original algorithm and an *isotropy* that sends it to another decomposition naturally with the same *tensor rank* but that does not involve complex coefficients anymore.

The next section is devoted to present the framework allowing this journey. Section 3 gives a *trilinear representation* of the tensor decomposition exhibited in this note. Furthermore, it presents the isotropy used to obtain it and the stabilizer of the considered orbits. Section 4 gives some straight line program that implements the presented theoretical algorithm.

2 Tensor decomposition encoding matrix product

In this note, we will use the representation of matrix multiplication algorithm given by a tensor decomposition of the corresponding bilinear map. For a more detailed discussion of this framework, we

refer the reader to [9]. Nevertheless, we expose it following the illustrative example of Strassen's matrix multiplication algorithm.

Illustrating example Consider three matrices:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}. \quad (1)$$

The product $\mathbf{C} = \mathbf{A} \cdot \mathbf{B}$ of 2×2 matrices can be obtained using Strassen's algorithm using the following computations:

$$\begin{aligned} \rho_1 &\leftarrow a_{11}(b_{12} - b_{22}), & \rho_4 &\leftarrow (a_{12} - a_{22})(b_{21} + b_{22}), \\ \rho_2 &\leftarrow (a_{11} + a_{12})b_{22}, & \rho_5 &\leftarrow (a_{11} + a_{22})(b_{11} + b_{22}), \\ \rho_3 &\leftarrow (a_{21} + a_{22})b_{11}, & \rho_7 &\leftarrow (a_{21} - a_{11})(b_{11} + b_{12}), \\ \rho_6 &\leftarrow a_{22}(b_{21} - b_{11}), & \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} &= \begin{bmatrix} \rho_5 + \rho_4 - \rho_2 + \rho_6 & \rho_2 + \rho_1 \\ \rho_5 + \rho_3 & \rho_5 + \rho_7 + \rho_1 - \rho_3 \end{bmatrix}. \end{aligned} \quad (2)$$

This straight-line program (a.k.a. SLP) encodes the following bilinear map over a field $\mathbb{R}$ with m, k, n — that are all equal to 2 in Strassen's algorithm:

$$\begin{aligned} \beta_{\text{MM}} : \mathbb{R}^{m \times k} \times \mathbb{R}^{k \times n} &\rightarrow \mathbb{R}^{m \times n}, \\ (\mathbf{A}, \mathbf{B}) &\mapsto \mathbf{A} \cdot \mathbf{B}. \end{aligned} \quad (3)$$

Indices m, k and n are used in this section for clarity, helping to easily distinguish the different spaces involved in the following sections such as $\mathbb{R}^{m \times k}$ the space of $m \times k$ matrices with coefficients in $\mathbb{R}$.

2.1 Matrix multiplication encoded by tensors decomposition

Following [9], we begin by expressing the bilinear map defining matrix product into the formalism of tensor decomposition, that provides a clear way to represent algorithms associated to such product and its associated symmetries.

Strassen's algorithm (2) is encoded as the decomposition of the matrix multiplication tensor, expressed as a sum of seven rank-one tensors $\mathcal{S}$ defined by the following relations:

$$\begin{aligned} \mathcal{S} = \sum_{i=1}^7 \mathbf{M}_i \otimes \mathbf{N}_i \otimes \mathbf{O}_i = & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ & + \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ & + \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \\ & + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \end{aligned} \quad (4)$$

in $(\mathbb{R}^{m \times k})^* \otimes (\mathbb{R}^{k \times n})^* \otimes \mathbb{R}^{m \times n}$ (with $m = k = n = 2$ in Equation (4)).

Definitions 2.1 Given $\mathcal{T}$ a tensor decomposition $\sum_{i=0}^r \mathbf{T}_{i1} \otimes \mathbf{T}_{i2} \otimes \mathbf{T}_{i3}$ where each summand is a rank-one tensor:

- the integer r is the tensor rank of tensor decomposition $\mathcal{T}$;
- the ordered list $[(\text{rank } \mathbf{T}_{ij})_{j=1 \dots 3}]_{i=1 \dots r}$ is called the type of tensor decomposition (rank $\mathbf{M}$ being the classical rank of the matrix $\mathbf{M}$). The type could be summarized as the multivariate polynomial $\sum_{i=0}^r X^{\text{rank } \mathbf{T}_{i1}} Y^{\text{rank } \mathbf{T}_{i2}} Z^{\text{rank } \mathbf{T}_{i3}}$.

In the above Strassen's tensor decomposition, the *tensor rank* is 7 and its type is $X^2 Y^2 Z^2 + 6XYZ$.

Furthermore, we are going to use in Section 2.2 and Section 3 the following relationship between the classical trace of linear operator and tensor representations introduced so far. The Frobenius inner product $\langle \cdot \rangle$ of a matrix $\mathbf{O}$ with a matrix product $\mathbf{M} \cdot \mathbf{N}$ defines a trilinear form $\text{Trace}(\mathbf{O}^\top \cdot \mathbf{M} \cdot \mathbf{N})$ as follows:

$$\begin{aligned} \mathcal{S}|_3 : \mathbb{R}^{m \times k} \times \mathbb{R}^{k \times n} \times (\mathbb{R}^{m \times n})^* &\rightarrow \mathbb{R}, \\ (\mathbf{M}, \mathbf{N}, \mathbf{O}^\top) &\mapsto \langle \mathbf{O}, \mathbf{M} \cdot \mathbf{N} \rangle. \end{aligned} \quad (5)$$

The complete contraction of this tensor $\mathcal{S}$ with an *input* tensor $\mathbf{A} \otimes \mathbf{B} \otimes \mathbf{C}$ is its *trilinear representation*:

$$(a_{11} + a_{22}) \otimes (b_{11} + b_{22}) \otimes (c_{11} + c_{22}) \quad (6)$$

$$+ (a_{12} - a_{22}) \otimes (b_{21} + b_{22}) \otimes c_{11} \quad (7)$$

$$+ (-a_{11} + a_{21}) \otimes (b_{11} + b_{12}) \otimes c_{22} \quad (8)$$

$$+ (a_{11} + a_{12}) \otimes b_{22} \otimes (-c_{11} + c_{12}) \quad (9)$$

$$+ a_{11} \otimes (b_{12} - b_{22}) \otimes (c_{12} + c_{22}) \quad (10)$$

$$+ a_{22} \otimes (-b_{11} + b_{21}) \otimes (c_{11} + c_{21}) \quad (11)$$

$$+ (a_{21} + a_{22}) \otimes b_{11} \otimes (c_{21} - c_{22}). \quad (12)$$

We present our result in Section 3 using this kind of representation.

2.2 Symmetries of matrix product tensor decomposition

Remark that the matrix product is associated to $\text{Trace}(\mathbf{A} \cdot \mathbf{B} \cdot \mathbf{C})$ by Equation (5) and that, given three invertible matrices $\mathbf{U}, \mathbf{V}, \mathbf{W}$ of suitable sizes and the classical trace properties, this trace is equal to:

$$\begin{aligned} \text{Trace}((\mathbf{A} \cdot \mathbf{B} \cdot \mathbf{C})^\top) &= \text{Trace}(\mathbf{C} \cdot \mathbf{A} \cdot \mathbf{B}) = \text{Trace}(\mathbf{B} \cdot \mathbf{C} \cdot \mathbf{A}) \\ \text{and to } \text{Trace}(\mathbf{U}^{-1} \cdot \mathbf{A} \cdot \mathbf{V} \cdot \mathbf{V}^{-1} \cdot \mathbf{B} \cdot \mathbf{W} \cdot \mathbf{W}^{-1} \cdot \mathbf{C} \cdot \mathbf{U}). \end{aligned} \quad (13)$$

These relations illustrate the next theorem and induce the isotropy action on matrix product tensor decomposition presented below:

Theorem 2.1 ([6, § 2.8]) *The isotropy group of the $m \times k$ by $k \times n$ matrix multiplication tensor is the semidirect product*

$$(\text{PSL}^\pm(\mathbb{R}^m) \times \text{PSL}^\pm(\mathbb{R}^k) \times \text{PSL}^\pm(\mathbb{R}^n)) \rtimes \mathfrak{S}_3, \quad (14)$$

where PSL stands for the group of matrices of determinant ± 1 and $\mathfrak{S}_3$ for the symmetric group on 3 elements.

As we do not harness the $\mathfrak{S}_3$ part of this isotropy group, we do not give further details on its action. Let us now describe how this group acts on tensor decompositions.

Definition 2.2 *Let $\mathcal{T}$ denotes a rank-one tensor $\mathbf{A} \otimes \mathbf{B} \otimes \mathbf{C}$ and let $\mathbf{g}$ denotes the isotropy associated to matrices $(\mathbf{U} \times \mathbf{V} \times \mathbf{W})$ in $\text{PSL}^\pm(\mathbb{R}^m) \times \text{PSL}^\pm(\mathbb{R}^k) \times \text{PSL}^\pm(\mathbb{R}^n)$; the action $\mathbf{g} \diamond \mathcal{T}$ of $\mathbf{g}$ on $\mathcal{T}$ is the rank-one tensor:*

$$(\mathbf{U}^{-\top} \cdot \mathbf{A} \cdot \mathbf{V}^\top) \otimes (\mathbf{V}^{-\top} \cdot \mathbf{B} \cdot \mathbf{W}^\top) \otimes (\mathbf{W}^{-\top} \cdot \mathbf{C} \cdot \mathbf{U}^\top). \quad (15)$$

This action is extended by additivity to higher tensor rank tensors. Given two isotropies $\mathbf{g}_1$ defined by matrices $(\mathbf{U}_1 \times \mathbf{V}_1 \times \mathbf{W}_1)$ and $\mathbf{g}_2$ defined by matrices $(\mathbf{U}_2 \times \mathbf{V}_2 \times \mathbf{W}_2)$ in $\text{PSL}^\pm(\mathbb{R}^m) \times \text{PSL}^\pm(\mathbb{R}^k) \times \text{PSL}^\pm(\mathbb{R}^n)$, the composition $\mathbf{g}_1 \circ \mathbf{g}_2$ is given by $(\mathbf{U}_1 \cdot \mathbf{U}_2 \times \mathbf{V}_1 \cdot \mathbf{V}_2 \times \mathbf{W}_1 \cdot \mathbf{W}_2)$.

The following theorem recalls that all 2×2 -matrix product algorithms with 7 coefficient multiplications are obtained by this single orbit of isotropies action on Strassen tensor decomposition:

Theorem 2.2 ([7, § 0.1]) *The group $\text{PSL}^\pm(\mathbb{R}^{2 \times 2})^{\times 3}$ acts transitively on the variety of fast algorithms multiplying 2×2 -matrices.*

More generally, isotropy action on any decomposition may define another matrix product algorithm of same tensor rank but with potentially more interesting characteristics. This fact is used in [3, 5] to increase the numerical accuracy of some fast matrix multiplication algorithms. There, we use the tensor decomposition orbit to express $\langle 2 \times 2 \times 2:7 \rangle$ and $\langle 3 \times 3 \times 6:40 \rangle$ with more suitable coefficients. Let us first do the opposite in the following example.

Example 2.1 *Consider the following isotropy:*

$$\mathbf{g} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (16)$$

The resulting trilinear representation associated to the tensor decomposition $\mathbf{g} \diamond \mathcal{S}$ is:

$$(a_{11} - Ia_{22}) \otimes (b_{11} + b_{22}) \otimes (c_{11} + Ic_{22}) \quad (17)$$

$$+ (a_{12} + Ia_{22}) \otimes (b_{21} + b_{22}) \otimes c_{11} \quad (18)$$

$$+ (-a_{11} - Ia_{21}) \otimes (b_{11} + b_{12}) \otimes c_{22}I \quad (19)$$

$$+ (a_{11} + a_{12}) \otimes b_{22} \otimes (-c_{11} + c_{12}) \quad (20)$$

$$+ a_{11} (b_{12} - b_{22}) \otimes (c_{12} + Ic_{22}) \quad (21)$$

$$- Ia_{22} (-b_{11} + b_{21}) \otimes (c_{11} + Ic_{21}) \quad (22)$$

$$+ (-Ia_{21} - Ia_{22}) \otimes b_{11} \otimes (Ic_{21} - Ic_{22}). \quad (23)$$

This last representation involves the complex number I but it is clear that there exists at least one point in the orbit of Strassen's tensor decomposition involving only integers.

So one can consider the following question:

Problem 2.1 *Is there a tensor decomposition in the orbit of $\langle 3 \times 4 \times 7 : 63 \rangle$ introduced in [10] that does not require complex numbers?*

We present now our main result in the following section.

3 A rational point in the orbit of AlphaEvolve's tensor decomposition

Using the Klein four stabilizer group (see [11]) of the original algorithm presented in [10], and educated guesses, we obtain the following isotropy:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} I & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} I & 0 & 0 & 0 & 0 & 0 \\ 0 & I & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & I \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (24)$$

that sends the original complex-valued tensor decomposition to the following rational one:

$$m_1 = (a_{21} + a_{24}) \otimes (b_{17} - b_{37} + b_{47}) \otimes \left(\frac{c_{21}}{2} - \frac{c_{22}}{2} + \frac{c_{24}}{2} + \frac{c_{26}}{2} + \frac{c_{27}}{2} \right), \quad (25)$$

$$m_2 = \left(a_{11} - a_{21} + \frac{a_{12}}{2} - \frac{a_{22}}{2} \right) \otimes \left(\frac{b_{32}}{2} + \frac{b_{42}}{2} - \frac{b_{12}}{2} - b_{22} - b_{14} + b_{15} + b_{17} \right) \otimes (c_{16} + c_{17} - c_{12}), \quad (26)$$

$$m_3 = \left(\frac{a_{32}}{2} - \frac{a_{22}}{2} + a_{24} - a_{34} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} + b_{43} + b_{46} - b_{47} \right) \otimes (c_{31} + c_{34} + c_{37}), \quad (27)$$

$$m_4 = (a_{11} + a_{13}) \otimes \left(b_{32} + b_{15} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{12} + c_{15} - c_{17}), \quad (28)$$

$$m_5 = \left(\frac{a_{14} + a_{34}}{2} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} + b_{43} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} + \frac{b_{44}}{2} + b_{46} - b_{47} \right) \otimes c_{34}, \quad (29)$$

$$m_6 = (a_{31} + a_{34}) \otimes \left(b_{11} + b_{46} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{31} + c_{36} + c_{37}), \quad (30)$$

$$m_7 = \left(a_{31} + \frac{a_{32}}{2} \right) \otimes (b_{16} - b_{36} - b_{46}) \otimes (-c_{16} + c_{26} + c_{36}), \quad (31)$$

$$m_8 = \left(a_{21} - a_{31} + \frac{a_{22}}{2} - \frac{a_{32}}{2} \right) \otimes \left(\frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17} \right) \otimes c_{26}, \quad (32)$$

$$m_9 = \left(\frac{a_{22}}{2} + \frac{a_{32}}{2} - a_{24} + a_{34} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} - \frac{b_{41}}{2} \right) \otimes (c_{21} + c_{31} + c_{34} + c_{37}), \quad (33)$$

$$m_{10} = \left(\frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{13} + a_{34} \right) \otimes \left(-b_{31} + b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{11} + c_{35}), \quad (34)$$

$$m_{11} = \left(\frac{a_{32}}{2} + a_{33} \right) \otimes (b_{13} - b_{33} + b_{43}) \otimes (c_{13} - c_{23} - c_{33}), \quad (35)$$

$$m_{12} = \left(\frac{a_{32}}{2} - \frac{a_{22}}{2} + a_{23} - a_{33} \right) \otimes \left(b_{31} + \frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} + b_{36} - b_{37} \right) \otimes (c_{33} - c_{35} + c_{37}), \quad (36)$$

$$m_{13} = \left(\frac{a_{32}}{2} + a_{34} \right) \otimes (b_{11} + b_{31} - b_{41}) \otimes (c_{11} - c_{21} - c_{31}), \quad (37)$$

$$m_{14} = \left(a_{21} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{13} \right) \otimes \left(b_{15} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (-c_{12} + c_{16} + c_{17} + c_{25}), \quad (38)$$

$$m_{15} = (a_{11} + a_{14}) \otimes \left(b_{42} - b_{14} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{12} - c_{14} - c_{17}), \quad (39)$$

$$m_{16} = \left(-a_{21} + a_{31} + \frac{a_{22}}{2} + \frac{a_{32}}{2} \right) \otimes \left(-\frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} \right) \otimes (c_{26} - c_{32} + c_{36} + c_{37}), \quad (40)$$

$$m_{17} = \left(\frac{a_{11} + a_{31}}{-\frac{a_{12}}{2} + \frac{a_{32}}{2}} \right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} \right) \otimes (c_{32} - c_{16}), \quad (41)$$

$$m_{18} = \left(\frac{\frac{a_{12}}{2} + \frac{a_{32}}{2}}{-a_{13} - a_{33}} \right) \otimes \left(b_{31} + \frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} - \frac{b_{15}}{2} + b_{25} - \frac{b_{35}}{2} - \frac{b_{45}}{2} + b_{36} - b_{37} \right) \otimes c_{35}, \quad (42)$$

$$m_{19} = \left(a_{11} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{23} \right) \otimes \left(b_{32} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{13} - c_{15} + c_{17} + c_{22}), \quad (43)$$

$$m_{20} = \left(a_{21} - a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2} \right) \otimes \left(b_{11} + b_{13} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17} \right) \otimes (c_{36} - c_{32} + c_{37}), \quad (44)$$

$$m_{21} = (a_{13} + a_{14}) \otimes \left(-b_{34} + b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (-c_{14} + c_{15} - c_{17}), \quad (45)$$

$$m_{22} = \left(\frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{33} - a_{14} \right) \otimes \left(b_{43} + b_{34} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (-c_{13} + c_{34}), \quad (46)$$

$$m_{23} = \left(-a_{11} + a_{21} + \frac{a_{12}}{2} - \frac{a_{22}}{2} \right) \otimes \left(\frac{b_{12}}{2} - b_{22} + \frac{b_{32}}{2} + \frac{b_{42}}{2} + b_{17} \right) \otimes c_{22}, \quad (47)$$

$$m_{24} = \left(-a_{11} + \frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{34} \right) \otimes \left(-b_{11} + b_{42} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{11} + c_{32}), \quad (48)$$

$$m_{25} = \left(a_{21} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{33} \right) \otimes \left(b_{13} - \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{23} + c_{32} - c_{36} - c_{37}), \quad (49)$$

$$m_{26} = \left(-\frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{13} + a_{24} \right) \otimes \left(b_{45} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{11} + c_{14} + c_{17} + c_{25}), \quad (50)$$

$$m_{27} = a_{32} \otimes \left(\frac{b_{41}}{2} - \frac{b_{13}}{2} - \frac{b_{11}}{2} - b_{21} - \frac{b_{31}}{2} - b_{23} + \frac{b_{33}}{2} - \frac{b_{43}}{2} + \frac{b_{16}}{2} - b_{26} - \frac{b_{36}}{2} - \frac{b_{46}}{2} + b_{27} \right) \otimes c_{37}, \quad (51)$$

$$m_{28} = \left(a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{23} \right) \otimes \left(b_{36} - \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{26} - c_{33} + c_{35} - c_{37}), \quad (52)$$

$$m_{29} = (a_{33} + a_{34}) \otimes \left(b_{31} + b_{43} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{31} + c_{33} + c_{37}), \quad (53)$$

$$m_{30} = \left(a_{21} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{14} \right) \otimes \left(-b_{14} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (-c_{12} + c_{16} + c_{17} - c_{24}), \quad (54)$$

$$m_{31} = \left(\frac{\frac{a_{12}}{2} + \frac{a_{32}}{2}}{+a_{13} + a_{33}} \right) \otimes \left(\frac{b_{13}}{2} + b_{23} + \frac{b_{33}}{2} + \frac{b_{43}}{2} + b_{34} - \frac{b_{15}}{2} - b_{32} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} - b_{37} \right) \otimes c_{13}, \quad (55)$$

$$m_{32} = \left(-a_{11} + a_{21} + \frac{a_{12}}{2} + \frac{a_{22}}{2} \right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} \right) \otimes (c_{12} - c_{16} - c_{17} + c_{22}), \quad (56)$$

$$m_{33} = \left(\frac{a_{22}}{2} - \frac{a_{32}}{2} + a_{23} - a_{33} \right) \otimes \left(\frac{b_{13}}{2} + b_{23} + \frac{b_{33}}{2} + \frac{b_{43}}{2} - b_{37} \right) \otimes c_{23}, \quad (57)$$

$$m_{34} = \left(\frac{a_{21}}{2} + \frac{a_{23}}{2} \right) \otimes (b_{17} + b_{37} - b_{47}) \otimes (-c_{22} + c_{23} - c_{25} + c_{26} + c_{27}), \quad (58)$$

$$m_{35} = \left(-a_{11} + \frac{a_{12}}{2} + \frac{a_{32}}{2} + a_{33} \right) \otimes \left(b_{32} - b_{13} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{13} + c_{32}), \quad (59)$$

$$m_{36} = \left(\frac{a_{12}}{2} + \frac{a_{22}}{2} - a_{13} + a_{23} \right) \otimes \left(-\frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} \right) \otimes (-c_{13} + c_{15} - c_{17} + c_{25}), \quad (60)$$

$$m_{37} = \left(\frac{a_{12}}{2} - \frac{a_{22}}{2} + a_{13} - a_{23} \right) \otimes \left(b_{32} - b_{34} + \frac{b_{15}}{2} - b_{25} - \frac{b_{35}}{2} + \frac{b_{45}}{2} + b_{37} \right) \otimes (c_{13} - c_{15} + c_{17}), \quad (61)$$

$$m_{38} = \left(\frac{a_{12}}{2} + \frac{a_{22}}{2} - a_{14} + a_{24} \right) \otimes \left(-\frac{b_{14}}{2} + b_{24} - \frac{b_{34}}{2} + \frac{b_{44}}{2} \right) \otimes (c_{11} + c_{14} + c_{17} + c_{24}), \quad (62)$$

$$m_{39} = \left(-\frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{33} + a_{24} \right) \otimes \left(-b_{43} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (-c_{23} + c_{31} + c_{34} + c_{37}), \quad (63)$$

$$m_{40} = a_{12} \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} - \frac{b_{44}}{2} - \frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} + b_{27} \right) \otimes c_{17}, \quad (64)$$

$$m_{41} = \left(-a_{11} + \frac{a_{12}}{2} \right) \otimes (-b_{12} + b_{32} + b_{42}) \otimes (c_{12} + c_{22} - c_{32}), \quad (65)$$

$$m_{42} = \left(\frac{\frac{a_{12}}{2} + \frac{a_{32}}{2}}{+a_{14} + a_{34}} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} + \frac{b_{41}}{2} - b_{42} + \frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} - \frac{b_{44}}{2} - b_{45} - b_{47} \right) \otimes c_{11}, \quad (66)$$

$$m_{43} = \left(\frac{a_{12}}{2} - a_{13} \right) \otimes (b_{15} - b_{35} + b_{45}) \otimes (c_{15} + c_{25} - c_{35}), \quad (67)$$

$$m_{44} = \left(\frac{a_{12}}{2} - a_{14} \right) \otimes (b_{14} + b_{34} - b_{44}) \otimes (c_{14} + c_{24} - c_{34}), \quad (68)$$

$$m_{45} = \left(\frac{-\frac{a_{12}}{2} + \frac{a_{32}}{2}}{a_{13} + a_{33}} \right) \otimes \left(\frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} - \frac{b_{15}}{2} + b_{25} + \frac{b_{35}}{2} - \frac{b_{45}}{2} \right) \otimes (-c_{13} + c_{35}), \quad (69)$$

$$m_{46} = \left(a_{31} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{24} \right) \otimes \left(-b_{46} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (-c_{26} + c_{31} + c_{34} + c_{37}), \quad (70)$$

$$m_{47} = \left(\frac{a_{22}}{2} + \frac{a_{32}}{2} - a_{23} + a_{33} \right) \otimes \left(\frac{b_{13}}{2} + b_{23} - \frac{b_{33}}{2} + \frac{b_{43}}{2} \right) \otimes (c_{23} + c_{33} - c_{35} + c_{37}), \quad (71)$$

$$m_{48} = \left(\frac{-a_{11} - a_{31}}{\frac{a_{12}}{2} + \frac{a_{32}}{2}} \right) \otimes \left(b_{11} - \frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + b_{13} - \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17} \right) \otimes c_{32}, \quad (72)$$

$$m_{49} = \left(-\frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{23} + a_{14} \right) \otimes \left(-b_{34} - \frac{b_{17}}{2} + \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{13} - c_{15} + c_{17} - c_{24}), \quad (73)$$

$$m_{50} = \left(a_{31} + \frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{14} \right) \otimes \left(b_{14} + b_{46} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (-c_{16} + c_{34}), \quad (74)$$

$$m_{51} = \left(-\frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{23} + a_{34} \right) \otimes \left(b_{31} + \frac{b_{17}}{2} - \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{21} - c_{33} + c_{35} - c_{37}), \quad (75)$$

$$m_{52} = \left(\frac{a_{22}}{2} - \frac{a_{32}}{2} + a_{24} - a_{34} \right) \otimes \left(\frac{b_{11}}{2} + b_{21} + \frac{b_{31}}{2} + \frac{b_{41}}{2} - b_{47} \right) \otimes c_{21}, \quad (76)$$

$$m_{53} = \left(\frac{a_{12}}{2} - \frac{a_{22}}{2} - a_{13} + a_{23} \right) \otimes \left(\frac{b_{15}}{2} - b_{25} + \frac{b_{35}}{2} + \frac{b_{45}}{2} + b_{37} \right) \otimes c_{25}, \quad (77)$$

$$m_{54} = \left(\frac{\frac{a_{32}}{2} - \frac{a_{12}}{2}}{+a_{14} + a_{34}} \right) \otimes \left(-\frac{b_{11}}{2} - b_{21} - \frac{b_{31}}{2} + \frac{b_{41}}{2} - \frac{b_{14}}{2} + b_{24} - \frac{b_{34}}{2} + \frac{b_{44}}{2} \right) \otimes (c_{11} + c_{34}), \quad (78)$$

$$m_{55} = \left(a_{11} - \frac{a_{12}}{2} + \frac{a_{22}}{2} + a_{24} \right) \otimes \left(b_{42} + \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{11} + c_{14} + c_{17} + c_{22}), \quad (79)$$

$$m_{56} = \left(\frac{a_{12}}{2} - \frac{a_{22}}{2} - a_{14} + a_{24} \right) \otimes \left(\frac{b_{14}}{2} - b_{24} + \frac{b_{34}}{2} + \frac{b_{44}}{2} - b_{47} \right) \otimes c_{24}, \quad (80)$$

$$m_{57} = \left(\frac{a_{23}}{2} + \frac{a_{24}}{2} \right) \otimes (-b_{17} + b_{37} + b_{47}) \otimes (c_{21} + c_{23} + c_{24} - c_{25} + c_{27}), \quad (81)$$

$$m_{58} = \left(a_{21} - \frac{a_{22}}{2} + \frac{a_{32}}{2} + a_{34} \right) \otimes \left(b_{11} - \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{21} + c_{32} - c_{36} - c_{37}), \quad (82)$$

$$m_{59} = \left(\frac{a_{11} + a_{31}}{+\frac{a_{12}}{2} + \frac{a_{32}}{2}} \right) \otimes \left(\frac{b_{12}}{2} + b_{22} - \frac{b_{32}}{2} - \frac{b_{42}}{2} + b_{14} - b_{15} + \frac{b_{16}}{2} + b_{26} + \frac{b_{36}}{2} + \frac{b_{46}}{2} - b_{17} \right) \otimes c_{16}, \quad (83)$$

$$m_{60} = \left(a_{31} + \frac{a_{12}}{2} + \frac{a_{32}}{2} - a_{13} \right) \otimes \left(b_{15} - b_{36} + \frac{b_{17}}{2} + \frac{b_{37}}{2} - \frac{b_{47}}{2} \right) \otimes (c_{16} + c_{35}), \quad (84)$$

$$m_{61} = (a_{31} + a_{33}) \otimes \left(b_{13} + b_{36} - \frac{b_{17}}{2} - \frac{b_{37}}{2} + \frac{b_{47}}{2} \right) \otimes (c_{33} + c_{36} + c_{37}), \quad (85)$$

$$m_{62} = \left(\frac{a_{12}}{2} - \frac{a_{22}}{2} + a_{14} - a_{24} \right) \otimes \left(b_{42} - \frac{b_{14}}{2} + b_{24} - \frac{b_{34}}{2} + \frac{b_{44}}{2} + b_{45} + b_{47} \right) \otimes (c_{11} + c_{14} + c_{17}), \quad (86)$$

$$m_{63} = a_{22} \otimes b_{27} \otimes c_{27}. \quad (87)$$

The type of this tensor decomposition is:

$$6X^2Y^3Z^2 + 15X^2Y^2Z^2 + 2XY^4Z + 6X^2YZ^2 + 12XY^3Z + 12XY^2Z + 10XYZ. \quad (88)$$

Several other representations of this algorithm are given in the following section, in appendix and are available in [11, 2].

4 Straight-line programs for the rational $\langle 3 \times 4 \times 7:63 \rangle$

We here give some (PLINOPT generated [2]) straight-line programs obtained from L, R, P matrices presented in Appendix B and associated to the tensor decomposition given by Equations (25) to (87):

Listing 1: SLP for input linear forms defined by matrix L

```
r12:=A12/2; 140:=r12-A11; r14:=A32/2; 10:=A21+A24; 16:=A31+r14; 110:=A33+r14;
122:=A21-A22/2+140; 126:=A32; 128:=A33+A34; 133:=(A21+A23)/2; 142:=r12-A13;
143:=r12-A14; t63:=140-122; t64:=A22-t63; t65:=128-110; 112:=t65+126; u67:=133*2;
t67:=u67-t64; t68:=10-t64; t69:=A12-142; t70:=t63+u67; t71:=t63+10; t72:=110-126;
116:=16-140; t74:=A12-143; 131:=122+A22; 115:=16+t63; 144:=110-142; 127:=16+t67;
12:=t68-t65; 19:=142+112; 113:=t64-142; 153:=112-143; 147:=126-116; 121:=110+143;
117:=142-t72; 111:=t67-t72; 151:=t71-112; 134:=110+140; 125:=t71-142; 152:=142+t67;
114:=t74-140; 137:=143+t71; 136:=t69-t70; 157:=112-t63; 135:=142+t70; 118:=u67-122;
11:=A12-131; 150:=t67+112; 123:=140+112; 138:=110+t68; 141:=112+t74; 161:=t74-t71;
14:=t65-143; 124:=110-t63; 159:=16+142; 17:=t64-16; 145:=16+t68; 155:=143+t68;
154:=10-122; 148:=t70-143; 149:=16+143; 132:=t70-110; 18:=112-t68; 15:=16+t65;
146:=110-t67; 160:=16+t72; 158:=A12+116; 130:=110+t69; 119:=126-115; 129:=t64-143;
13:=t69-140; 120:=t69-143; 156:=133+t63+(10-A22)/2; 139:=A12; 162:=A22;
```

Listing 2: SLP for input linear forms defined by matrix R

```
y28:=B17-B37; r56:=B47-y28; r0:=B47+y28; r33:=B17+B37-B47; y33:=B14+B34; y34:=B13+B43;
y35:=B11+B31; y36:=B32+B42; y37:=B36+B46; y38:=B15+B45; y39:=B24-y33/2; y40:=B21+y35/2;
y41:=B25-y38/2; y42:=B22-y36/2; y43:=B26+y37/2; y44:=B23+y34/2; b45:=B33/2;
r46:=y44-b45; b46:=B35/2; r35:=y41+b46; b47:=B41/2; r8:=y40-b47; b48:=B12/2;
r31:=y42+b48; b49:=B44/2; r37:=y39+b49; b50:=B16/2; r15:=y43-b50; b51:=r56/2;
y51:=b51-B34; b52:=r0/2; r45:=b52-B46; r50:=B31-b51; r54:=B42+b52; b55:=r33/2;
r18:=B32+b55; r27:=B36-b55; r32:=y44+b45-B37; y64:=B42+B45+r37; y68:=y41-b46;
r51:=b47+y40-B47; y72:=r35+B34-B32; y74:=y42-b48; y75:=B46+B43+r8; r7:=b50+y43-B17;
r19:=B11+B13+r15-B17; y78:=B15-B14-r31; r55:=b49-B47-y39; r11:=r46+B36+B31-B37;
r1:=B17+y78; r2:=y75-B47; r3:=B15+r18; r4:=y75+r55; r5:=B11-r45; r6:=B16-y37;
r9:=B45-r50; r10:=y34-B33; r12:=y35-B41; r13:=B15+b55; r14:=r54-B14; r16:=r31+r15;
r17:=y68+r11; r20:=B45+y51; r21:=B43-y51; r22:=B17-y74; r23:=r54-B11; r24:=B13-b55;
r25:=B45+b51; r26:=B27-r46-r8-r15; r28:=B43+r50; r29:=b52-B14; r30:=r32+y72;
r34:=r18-B13; r36:=B37-y72; r38:=b51-B43; r39:=B27+r35+r31-r37; r40:=y36-B12;
r41:=r51-y64; r42:=y38-B35; r43:=y33-B44; r44:=r46+r35; r47:=y74+r19; r48:=y51;
r49:=B14-r45; r52:=B37-y68; r53:=r37-r8; r57:=B11-b52; r58:=r7-y78; r59:=B15-r27;
r60:=B13+r27; r61:=B47+y64; r62:=B27;
```

Listing 3: products

```
p0:=10*r0; p1:=11*r1; p2:=12*r2; p3:=13*r3; p4:=14*r4; p5:=15*r5; p6:=16*r6; p7:=17*r7;
p8:=18*r8; p9:=19*r9; p10:=110*r10; p11:=111*r11; p12:=112*r12; p13:=113*r13;
p14:=114*r14; p15:=115*r15; p16:=116*r16; p17:=117*r17; p18:=118*r18; p19:=119*r19;
p20:=120*r20; p21:=121*r21; p22:=122*r22; p23:=123*r23; p24:=124*r24; p25:=125*r25;
p26:=126*r26; p27:=127*r27; p28:=128*r28; p29:=129*r29; p30:=130*r30; p31:=131*r31;
p32:=132*r32; p33:=133*r33; p34:=134*r34; p35:=135*r35; p36:=136*r36; p37:=137*r37;
p38:=138*r38; p39:=139*r39; p40:=140*r40; p41:=141*r41; p42:=142*r42; p43:=143*r43;
p44:=144*r44; p45:=145*r45; p46:=146*r46; p47:=147*r47; p48:=148*r48; p49:=149*r49;
p50:=150*r50; p51:=151*r51; p52:=152*r52; p53:=153*r53; p54:=154*r54; p55:=155*r55;
p56:=156*r56; p57:=157*r57; p58:=158*r58; p59:=159*r59; p60:=160*r60; p61:=161*r61;
p62:=162*r62;
```


Listing 4: SLP for input linear forms defined by matrix P

```

q12:=p56+p33; q20:=p61+p54+p37+p25; q29:=p48+p36-p35+p18; q34:=p57+p24-p19-p15;
q40:=p50-p46+p27-p11; q53:=p45+p38+p8+p2; q54:=p29-p31+p13+p1; z37:=p0/2; z36:=p56+z37;
q55:=p33+z37; C24:=p55-p48+p43+p37-p29+z36; C21:=p57+p51+p50-p12+p8+z36;
C25:=p52+p42+p35+p25+p13-q12; C26:=p27-p45+p15+p7+p6+q55; C22:=p54+p40+p31+p22+p18-q55;
C27:=p62+z37+q12; C23:=p46-p38+p32+p24-p10+q12; z23:=p5+q53;
C34:=p53+p49-p43+p21+p4+q53; q56:=q20-p20-p14; C11:=p53+p41+p23+p12+p9+q20;
q57:=p60+p28-q40; C35:=p59+p44-p42+p17+p9+q40; C32:=p47-p40+p34+p23+p16+q34;
q58:=q54-p3; C16:=p59+p58-p49-p16-p6+q54; C36:=p60+p6+p5-q34; C15:=p42+p20+p3-q29;
C13:=p34-p44+p30-p21+p10+q29; C12:=p40+p14-q58; C31:=p28-p12+z23; C14:=p43+q56;
C17:=p39+q29+q58+q56; C33:=q57-p10; C37:=p26-q34+z23+q57;

```

These straight-line programs require:

- 73 additions and 6 multiplications (binary shifts) for L;
- 98 additions and 15 multiplications (binary shifts) for R;
- 107 additions and 1 multiplication (binary shifts) for P.

5 Concluding remarks

This note concludes successfully a work started in [4] whose purpose is to simplify the complex-valued fast matrix multiplication algorithms introduced in [10].

As done in [4] for the rational $\langle 4 \times 4 \times 4 : 48 \rangle$ tensor decomposition, we could explicit the leading constant of asymptotic complexity of the $\langle 3 \times 4 \times 7 : 63 \rangle$ tensor decomposition presented in this note. This would use the number of operations of the associated SLPs, given in Section 4. But, to do so we have to consider square matrices, and the natural candidate to such a study should be the tensor decomposition $\langle 84 \times 84 \times 84 : 250057 \rangle$.

Now, as 84 is equal to $3 \cdot 4 \cdot 7$ and 250057 is equal to 63^3 , such a tensor decomposition is obtained by the tensor product of $\langle 3 \times 4 \times 7 : 63 \rangle$ with $\langle 7 \times 3 \times 4 : 63 \rangle$ and $\langle 4 \times 7 \times 3 : 63 \rangle$. Up to Tellegen's transposition principle, the other two tensor decompositions would share the same L, R, P matrices (permuted and transposed) and thus similar SLPs. This would then give an algorithm with cost $O(n^{3 \log_{84} 63})$, that is $O(n^{2.805217355})$.

Nevertheless, as shown in [1], there exists a tensor decomposition $\langle 84 \times 84 \times 84 : 226100 \rangle$ with a better asymptotic cost: $O(n^{\log_{84} 226100})$ that is $O(n^{2.782496596})$.

References

- [1] Charles-Éric Drevet, Md. Nazrul Islam, and Éric Schost. Optimization techniques for small matrix multiplication. *Theoretical Computer Science*, 412(22):2219–2236, May 2011. doi:10.1016/j.tcs.2010.12.012. (cited in sec. 8)
- [2] Jean-Guillaume Dumas, Bruno Grenet, Clément Pernet, and Alexandre Sedoglavic. PLinOpt, a collection of C++ routines handling linear & bilinear programs, January 2024. 8.09 kSLOC. URL: <https://github.com/jgdumas/plinopt>. (cited in sec. 7), (cited in sec. 9)
- [3] Jean-Guillaume Dumas, Clément Pernet, and Alexandre Sedoglavic. Strassen's algorithm is not optimally accurate. In Shaoshi Chen, editor, *ISSAC'2024, Proceedings of the 2024 International Symposium on Symbolic and Algebraic Computation*, Raleigh, NC, USA, pages 254–263, New York, July 2024. ACM Press. URL: <https://hal.science/hal-04441653>, doi:10.1145/3666000.3669697. (cited in sec. 3)
- [4] Jean-Guillaume Dumas, Clément Pernet, and Alexandre Sedoglavic. A non-commutative algorithm for multiplying 4×4 matrices using 48 non-complex multiplications. Technical Report 2506.13242, arXiv, June 2025. arXiv:2506.13242. (cited in sec. 8)
- [5] Jean-Guillaume Dumas, Clément Pernet, and Alexandre Sedoglavic. Towards automated generation of fast and accurate algorithms for recursive matrix multiplication. Technical report, HAL open science, March 2025. URL: <https://hal.science/hal-04995684>. (cited in sec. 3)

- [6] Hans Friedrich Groote, de. On varieties of optimal algorithms for the computation of bilinear mappings I. The isotropy group of a bilinear mapping. *Theoretical Computer Science*, 7(2):1–24, 1978. doi:10.1016/0304-3975(78)90038-5. (cited in sec. 3)
- [7] Hans Friedrich Groote, de. On varieties of optimal algorithms for the computation of bilinear mappings II. Optimal algorithms for 2×2 -matrix multiplication. *Theoretical Computer Science*, 7(2):127–148, 1978. doi:10.1016/0304-3975(78)90045-2. (cited in sec. 3)
- [8] Elaye Karstadt and Oded Schwartz. Matrix multiplication, a little faster. In Christian Scheideler and Mohammad Hajiaghayi, editors, *SPAA ’17: Proceedings of the 29th ACM Symposium on Parallelism in Algorithms and Architectures*, pages 101–110, New York, NY, USA, July 24–26 2017. ACM Press. doi:10.1145/3087556.3087579. (cited in sec. 11)
- [9] Joseph M. Landsberg. *Geometry and complexity theory*, volume 169 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, December 2016. doi:10.1017/9781108183192. (cited in sec. 2)
- [10] Alexander Novikov, Ngàn Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, Adam Zsolt Wagner, Sergey Shirobokov, Borislav Kozlovskii, Francisco J. R. Ruiz, Abbas Mehrabian, M. Pawan Kumar, Abigail See, Swarat Chaudhuri, George Holland, Alex Davies, Sebastian Nowozin, Pushmeet Kohli, and Matej Balog. AlphaEvolve: A coding agent for scientific and algorithmic discovery. Technical Report 2506.13131, arXiv, May 2025. URL: <https://arxiv.org/abs/2506.13131>. (cited in sec. 1), (cited in sec. 4), (cited in sec. 8)
- [11] Alexandre Sedoglavic. Fast matrix multiplication database [online]. December 2017. URL: <http://fmm.univ-lille.fr/>. (cited in sec. 4), (cited in sec. 7)
- [12] Alexey Vladimirovich Smirnov. Several bilinear algorithms for matrix multiplication problems $\langle 3, p, q \rangle$. Technical report, April 2021. URL: [researchgate:350897049](https://researchgate.net/publication/350897049). (cited in sec. 1)
- [13] Volker Strassen. Gaussian elimination is not optimal. *Numerische Mathematik*, 13(4):354–356, August 1969. doi:10.1007/BF02165411. (cited in sec. 1)

A Source codes

All the different matrices presented in this appendix can be found in the **PLINOPT** library’s data directory [2]. For straight line programs presented in Appendix B:

- [data/3x4x7_63_rational_L.sms](#);
- [data/3x4x7_63_rational_R.sms](#);
- [data/3x4x7_63_rational_P.sms](#).

For alternative basis based straight line programs presented in Appendix C:

- [data/3x4x7_63_rational-ALT_L.sms](#), [data/3x4x7_63_rational-CoB_L.sms](#);
- [data/3x4x7_63_rational-ALT_R.sms](#), [data/3x4x7_63_rational-CoB_R.sms](#);
- [data/3x4x7_63_rational-ALT_P.sms](#), [data/3x4x7_63_rational-CoB_P.sms](#).

B LRP representation of the algorithm

Notation B.1 Given an $m \times k$ -matrix A , we denote by $A_{i,*}$ the i th row and by $\text{Vect}(A)$ the row-major vectorization of this matrix, i.e. the vector v in $\mathbb{R}^{mk}$ of the matrix coefficients such that $v_{ik+j} = a_{i,j}$.

Definition B.1 The LRP representation of a tensor decomposition $T = \sum_{i=1}^r M_i \otimes N_i \otimes O_i$ in the tensor product space $(\mathbb{R}^{m \times k})^* \otimes (\mathbb{R}^{k \times n})^* \otimes \mathbb{R}^{m \times n}$, is formed by the three matrices L in $\mathbb{R}^{r \times mk}$, R in $\mathbb{R}^{r \times kn}$, and P in $\mathbb{R}^{mn \times r}$, such that $L_{i,*}$ is $\text{Vect}(M_i)$, $R_{i,*}$ is $\text{Vect}(N_i)$ and $P_{*,i}$ is $\text{Vect}(O_i^\top)$.

[illegible]